Point Hub

Our own scoreboard. 1.67 times the size of the Tech4Sport box, a small score panel facing each bench, a front that tilts toward the first referee, a touch screen for the scorer in a flap at the back, all the ports on one removable plate. 32 steps.



SIZE 102 × 38 × 22 cm	WEIGHT about 8.5 kg
MAIN SCREEN 192 × 64 LEDs, 5 mm	BENCH PANELS 2 × 32 × 64 LEDs
FRONT TILT 0–20°, five notches	SCORER SCREEN 10.1″ touch, in a flap
POWER USB-C charger, 65 W+	PORTS POWER, DATA, BACKUP (USB-A + USB-C)

Sizes in the steps still depend on the exact parts that arrive. Marked “open point”: decided at the dry fit. Marked “check before you rely on it”: not tested yet.

A First light 01–01
B Worksho 02–05
C Case 06–16
D Screens 17–19
E Electroni 20–28
F Finish 29–32

In the box

The wood letters match the cut list and the drawings.

CUT LIST · JUMBO CUT-TO-SIZE, OECOPLAN POPLAR 6 MM



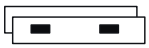
A Front frame ×1

1004 × 366; window 960 × 320 cut at home. Tilts on the piano hinge



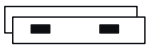
B Back ×1

1020 × 380; flap opening 280 × 205, port plate hole, horn hole, vents at home



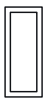
C1 Top ×1

1020 × 214; two handle cut-outs 129 × 52 at home



C2 Bottom ×1

1020 × 214; stand cup and thread holes after the dry fit



D Ends ×2

368 × 214; window 160 × 320 each, for the bench screens



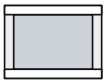
F Flap ×2

260 × 180, two layers glued (12 mm); trim to the screen



P1 Pocket back ×1

292 × 217



P2 · P3 Pocket walls ×1

blank 292 × 150 → 2 × 292 × 25 and 2 × 205 × 25



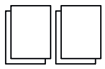
K Bottom cap ×1

blank 1004 × 150; final depth about 140, set at the dry fit



H Cheeks ×1

blank 370 × 150 → 370 × 130 → two triangles



Y Backing plates ×1

blank 300 × 208 → 2 × 208 × 140, glued: 12 mm



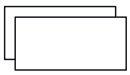
T Pull-out foot ×1

600 × 150



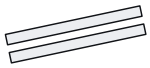
Spare ×1

400 × 300, for practice cuts and remakes



Transport covers, 3 mm ×4

poplar 3 mm: front 1020 × 380, ends 380 × 220 ×2, lid 300 × 225



Beech battens 15 × 20 ×5

× 1000: 2 long battens, 2 foot rails, 2 tray posts 362, blocks, 1 spare. 15 × 15 is not stocked in Zürich

FRONT AND TILT



Smoked polycarbonate 3 mm ×3

Makrolon UV 2760 grey (about 62 % light): 990 × 350 + 2 × 172 × 332



LED module P5 64 × 32 ×10

indoor, ICN2037BP + 74HC138, one batch: 8 + 2 spare



M3 magnetic screws ×40

size 1313, four per module



Steel flat bar 20 × 4 ×3

cut to 3 × 360 (must be steel: the magnets hold on it); two 1 m bars



Piano hinge

about 1000 × 32, cut to 1000



Cap hinges ×3

K to A's bottom edge: 3 butt hinges 50 mm, 3 × 12 screws, a 6 mm doubler under each leaf



Ratchet stays ×2

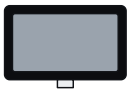
aluminium flat 25 × 4 (or steel), 200 long, five notches



M6 × 30 hex bolt ×2

with nylon lock nut: the pin each stay drops onto

SCREEN FLAP



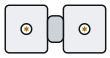
10.1″ DSI touch screen

Waveshare 10.1″ DSI LCD (C), 1280 × 800



Display cable

standard–mini DSI, 300 mm (Raspberry Pi SC1132)



Torque hinges ×2

friction hinge 1.2 N·m



Magnetic catches ×6

Hettich 27 × 18, screw-on, 2 kg, adjustable (flap and back)

CASE



K16 recessed handles ×2

spring-back, rim 155 × 83.3, cut-out 129 × 52, + 6 × M4 × 16 pan Torx A2 and lock nuts each



Rubber pads ×8

clear, self-adhesive, 20 × 20: 4 under the box, 2 under the foot, 2 spare



Corner caps ×4

black rubber, front corners only, trimmed to the ends' edge



Turn buttons ×12

bench screens, back panel, foot



Stand cup

35 mm (K&M 195/8 flange), M4 screws



Tripod threads ×2

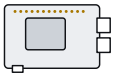
¼″-20 and ⅜″-16 wood inserts, in the 12 mm plate



Splash protection

black silicone, EPDM D-seal, brush seal, dust caps for all four ports, vent mesh; optional 2K matte clear

ELECTRONICS



Raspberry Pi 5, 8 GB

+ Active Cooler (SC1148). Fallback if Pi 5 + Piomatter fails: Pi 4 2 GB with the hzeller library



LED driver board

Adafruit Triple LED Matrix Bonnet (6358)



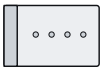
Stacking header

40 pin, tall: reaches pins 27/28 under the bonnet



microSD card ×2

32 GB, prepared; one spare



Converter

Mean Well RSD-60G-5, 9–36 V in, 5 V 12 A out



USB-C power board

STUSB4500 (I²C 0x28)



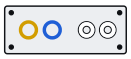
USB-C panel sockets ×2

panel-mount extensions, 30 cm: POWER and DATA



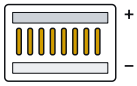
BACKUP cables ×2

USB-A 3.0 panel extension + USB-A plug to USB-C panel socket, about 30 cm: for a stick or SD reader



Port plate

2–3 mm black plastic, blank about 160 × 80; aluminium only after the meter check (BUILD.md §8.9)



Fuse box

12-way ATO, 5 A per way + spares



INA226 current sensors ×3

R002 (2 mΩ) boards: 0x40 on the 5 V out, 0x41 on the input (A0 bridged to VCC), one spare



MCP23017 module

I²C 0x20: switches the horn



MOSFET module

AOD4184, isolated (optocoupler input)



Horn buzzer

SFM-27 active piezo, continuous (not "-II"), 3–24 V, about 95 dB



Status lights ×2

STATUS and LINK: RGB 5 mm matte common-cathode LEDs in metal panel holders, Ø8 holes; resistors 150 / 47 / 47 Ω



Power button

16 mm, self-reset, with a green 6 V ring: the ring is the POWER light. Ø16 hole



Clock + temperature ×2

DS3231 (0x68) with CR2032, SHT40 (0x44)



Tablet charging module

USB-C PD 65 W (DMyond), off the fused input through 3 A



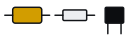
USB-C hub

Ethernet (RTL8153) + charging input: the DATA cable to the Pi



Electronics tray

2 mm aluminium, up to 400 × 200: everything mounts on it; stands on two batten posts



Small parts

TVS P6KE27A, 5 A and 3 A inline fuses, 10 kΩ resistor

TRANSPORT



Magnets N48 ×20

Ø8 × 3, sunk into the box borders: 8 front, 4 per end, 4 lid



M5 body washers ×20

sunk about 1 mm into the covers, opposite each magnet



Felt

dark grey, 0.8–1 mm (not 4 mm), inside the covers



USB-C cable, 3 m

PD + data, for speaker-stand set-ups: the tablet stays on the table



Custom bag

after the build, AliExpress: 104 × 40 × 24 cm inside, 15–20 mm padding, 600D nylon, zip on 3 sides, handles + strap, outer pocket

SCREWS, GLUE, PAINT



Wood screws T20 ×70

3.5 × 16 (shell, blocks, back); 3.5 × 30 for the two stay pivots



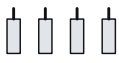
Machine screws T20 ×4

M4 × 16 with nuts: stand flange



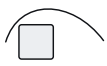
Port plate screws ×4

M4 × 20 with lock nuts: M4 × 16 is too short here



Standoffs ×16

M2.5, about 10–11 mm: Pi, bonnet, small boards and the hub board



Cable-tie mounts ×10

self-adhesive, with small ties: leads flat on the tray



Wood glue D3

water-resistant PVA (UHU Holzleim wasserfest)



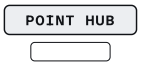
Paint

matte black, gloss black for the front face, filler, P120 + P180



Wire, red + black

2.5 mm² · 1 mm² · ferrules · heat-shrink

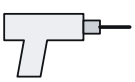


Lettering and label ×2

“POINT HUB” front, back label: after first light

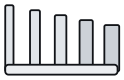
Tools

The kit we bought; the workshop has the rest.



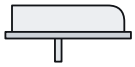
Drill/driver

Bosch GSB 18V-65, T20 bits



Drill bits

wood bits and HSS bits (Ø2, 2.5, 2.7, 3.2, 4.5, 4.8, 6.5, 8)



Jigsaw

Bosch GST 18V-125 S



Jigsaw blades

T101B straight wood · T101AOF curves · T118B aluminium/steel · T118A piano hinge



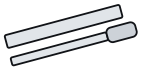
Hole saws

Wolfcraft Ø32 and Ø45



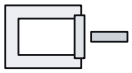
Centre punch

automatic



Files

flat set + needle files



Clamps

4 × Wolfcraft EHZ Easy 300

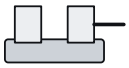
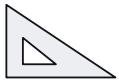


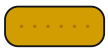
Table vise

50 mm



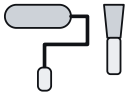
Squares

speed square (angles for the tilt) + square set



Sanding

sponges + paper P120 / P180



Roller and brush

10 cm foam roller, 25 mm brush



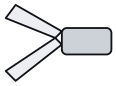
Painter's tape

still to buy: holds the long joints



Soldering iron

Pinecil V2 (USB-C PD 20 V), Sn63/Pb37 0.8 mm, tip cleaner



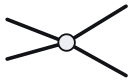
Ferrule crimper

HSC8 6-4A + 400 ferrules



Heat-shrink

2:1 assortment



Wire stripper

automatic, 0.2–6 mm² (Jokari Super 4 Plus or Knipex)



Side cutter

140–160 mm



Forstner bits

Ø8 for the magnets, washer size for the covers; not spade bits



Multimeter

UNI-T UT33B+

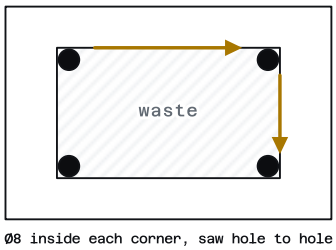


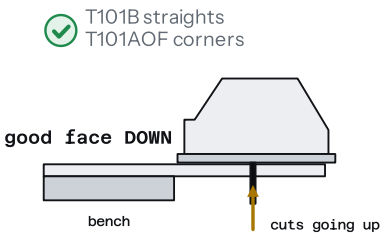
Laser meter

Bosch GLM 50-27 C

How to do it

Each technique once, with a drawing. The steps link back here. The links open guides and videos from other sites.





Clamp the sheet so the blade runs clear of the bench.

Don't push hard or twist the blade: let it cut.

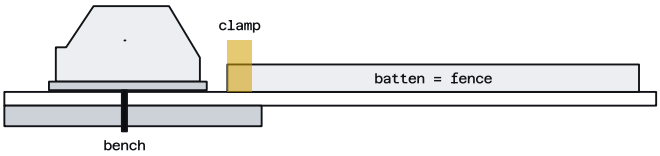
Cut a window with the jigsaw

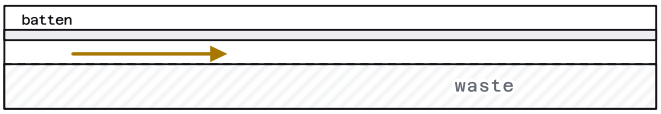
1. Take the battery out, fit the blade, push it till it locks. T101B for straights, T101AOF for corners.
2. Mark the cut, with masking tape on the line, and lay the good face **down**: the blade cuts on the up stroke.
3. Clamp the sheet so the blade has free space below it.
4. Drill Ø8 inside every corner.
5. Set orbit to 0, for the cleanest cut.
6. Saw from hole to hole, with the shoe flat, letting the blade do the work.
7. File the corners clean.

Bosch: cutting an opening, drill the corners first (Norwegian, good pictures) bosch-diy.com

Bosch GST 18V-125 S: product page, with the manual bosch-professional.com

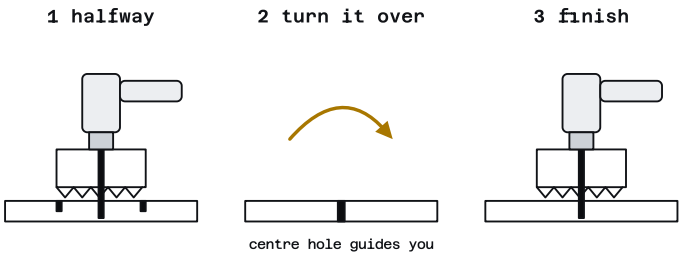






Saw a straight line

1. Measure from the blade to the edge of the shoe, on an offcut.
2. Clamp a straight batten that far from your line.
3. Keep the shoe tight against the batten for the whole cut.
4. Cut a little wide and sand down to size.



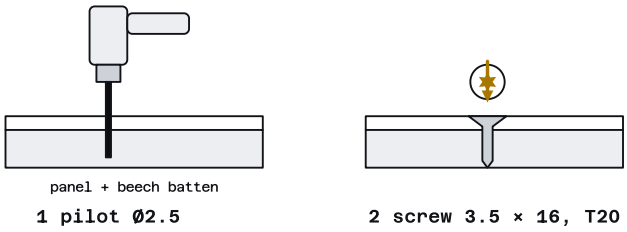
- Drill in gear 1 (slow), hammer off. Short bursts, clear the dust.
- Plug stuck? Lever it out through the slots with a screwdriver.

Cut a round hole

1. Mark the centre and clamp the board.
2. Drill in gear 1, hammer off.
3. Stop halfway, when the centre bit has come through.
4. Turn the board over and finish from the other side, using the same centre hole.

This Old House: how to use a hole saw
thisoldhouse.com

Ron Hazelton: clean holes with a hole saw
ronhazelton.com

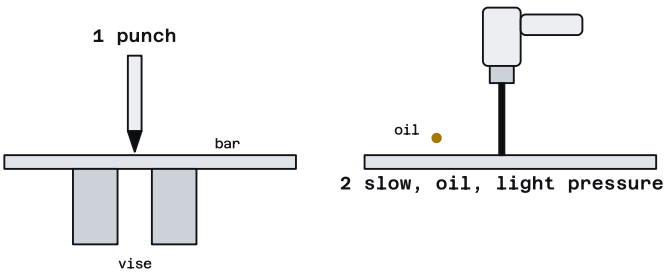


- Beech splits without a pilot hole. So does plywood near an edge.
- Clutch ring low, gear 1: stop when the head sits flush.
- Hammer mode off, always: it wrecks plywood.

Pre-drill and screw

1. Drill Ø2.5 through the panel and into the beech.
2. Set the drill's clutch low, gear 1, T20 bit.
3. Drive the screw until the head sits flush.

Bosch GSB 18V-65: product page, with the manual
bosch-professional.com



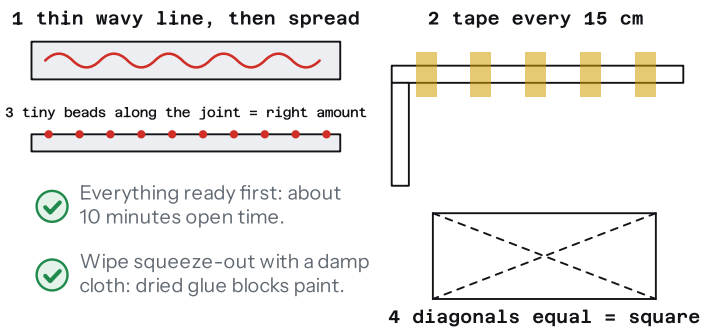
- Jigsaw: T118B, low speed, orbit 0. Clamp the bar close to the cut.
- File every edge and hole: burrs cut wires and fingers. Wear glasses.

Cut and drill metal

1. Mark the spot and press the centre punch on it.
2. Clamp the bar in the vise.
3. Put a drop of oil on the mark.
4. Drill slowly with an HSS bit, gear 1, light pressure.
5. Saw with the T118B blade at low speed.
6. File every edge and hole. Wear glasses.

Instructables: how to use a centre punch
instructables.com

Tivoly: drilling guide tivoly.com



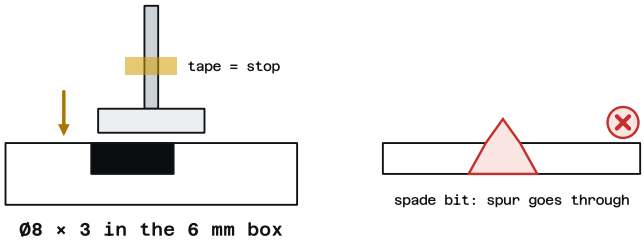
- Everything ready first: about 10 minutes open time.
- Wipe squeeze-out with a damp cloth: dried glue blocks paint.

Glue and tape a joint

1. Dry fit first. Lay out parts, tape, screws and a damp cloth.
2. Run a thin line of D3 glue and spread it.
3. Join the parts: tiny beads along the joint mean the right amount.
4. Tape every 15 cm across the joint, pulled tight. Clamp the short joints.
5. Check the diagonals are equal.
6. Wipe off the squeeze-out and let it dry overnight.

Fine Woodworking: tips on gluing `finewoodworking.com`

Rockler: how long to clamp `rockler.com`



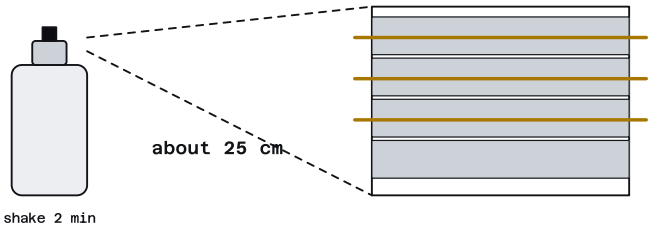
- Forstner bit, gear 1, slow. Try it on an offcut, then check the depth.
- A magnet should sit flush, not proud. Glue it in after the paint.

Sink a magnet or washer pocket

1. Wrap tape round the bit at the depth you want.
2. Try it on an offcut and check with the magnet.
3. Drill slowly, gear 1, stopping when the tape touches.
4. For the 3 mm covers, drill 1 mm deep only.

Woodcraft: depth stops for Forstner bits `woodcraft.com`

KM Tools: stop Forstner tear-out `kmtools.com`



- Start and stop past the edge. Thin coats, 50 % overlap. Test on an offcut.
- Dry, dust-free, right temperature (see can). P180 between coats.
- 2K paint: only outside, with an A2 respirator and gloves.

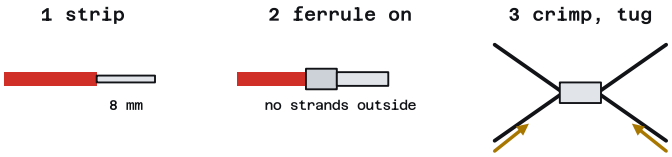
Spray paint

1. Sand smooth, remove the dust, mask what must stay bare.
2. Shake the can for as long as it says.
3. Try it on an offcut first.
4. Spray from about 25 cm in long passes. Start and stop past the edge.
5. Thin coats. Wait as the can says, then sand lightly with P180.
6. 2K paint: outside only, A2 respirator and gloves.

Instructables: perfectly spray paint instructables.com

Homebuilding: spray painting wood homebuilding.co.uk

Krylon: how to spray paint krylon.com



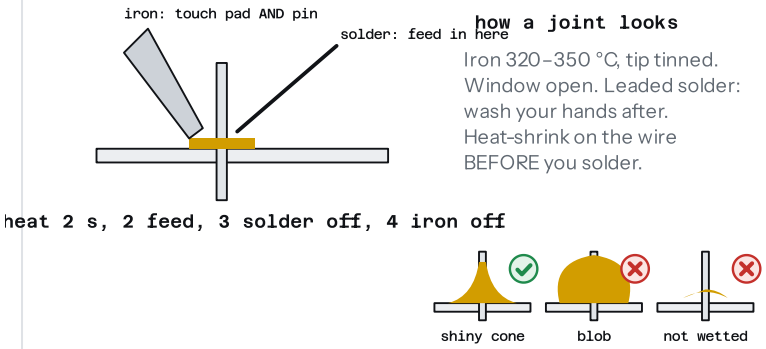
- Twist the strands, then push them right to the ferrule's end.
- Size: pick the ferrule by the wire size, and the matching crimper hole.
- Tug hard: it must not move. If it does, cut it off and use a new one.
- Ferrules are single use, and for stranded wire only.

Strip and crimp a wire end

1. Strip 8 mm: automatic stripper, one squeeze.
2. Twist the strands tight.
3. Push the ferrule on: no strands left outside.
4. Crimp in the HSC8 until it opens by itself.
5. Tug hard. If it moves, cut it off and use a new one.

DigiKey: using and crimping wire ferrules forum.digikey.com

Herth+Buss: crimping video herthundbuss.com



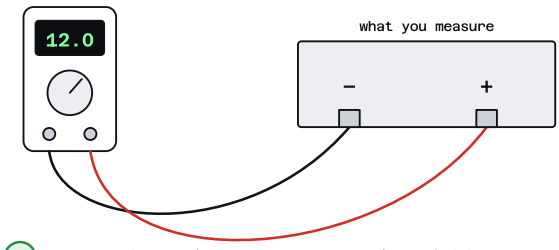
Solder a joint

1. Power the Pinecil from a USB-C PD charger, 65 W. Set 320–350 °C and let it heat up.
2. Tin the tip: touch solder on it, wipe it on the cleaner.
3. Slide heat-shrink on the wire now, if it needs some.
4. Hold the tip on the pad and the pin together for about 2 s.
5. Feed solder into the joint, not onto the iron.
6. Take the solder away, then the iron.
7. Look: shiny and cone-shaped is right. Slide the heat-shrink over and warm it evenly.

Adafruit: guide to excellent soldering
learn.adafruit.com

Adafruit: what bad joints look like
learn.adafruit.com

PINE64: Pinecil wiki wiki.pine64.org



- Dial on DC volts (V with a straight line). Black COM, red V.
- A minus sign means red and black are swapped: nothing breaks.

Measure a voltage

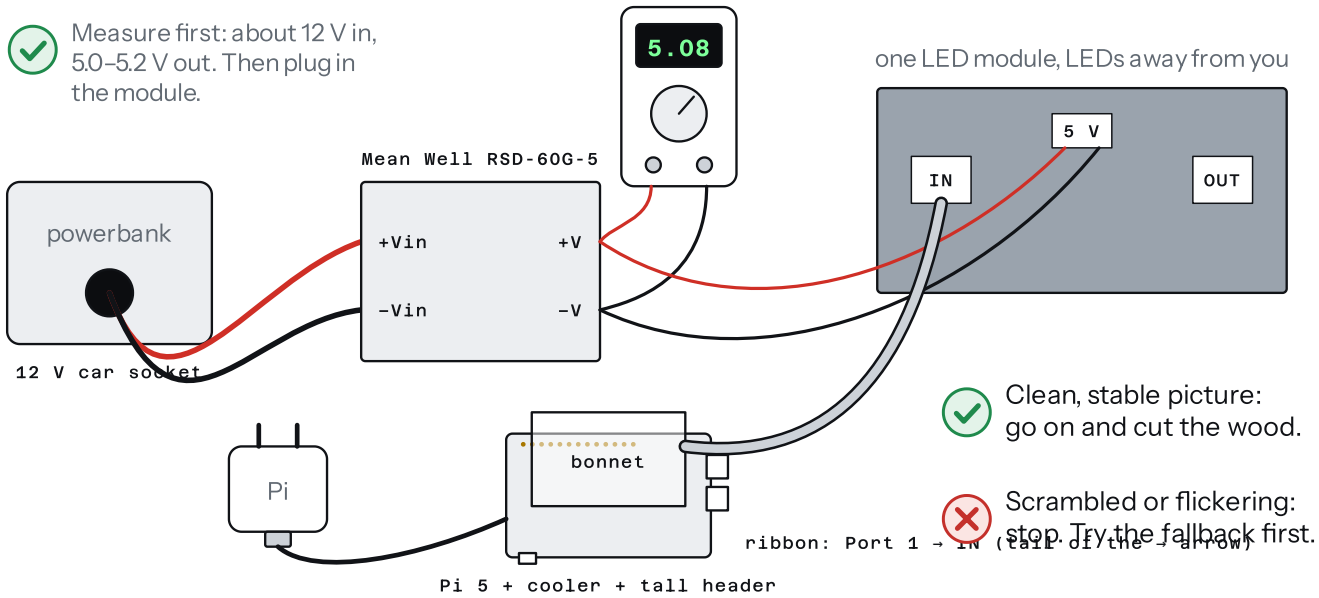
1. Black lead in COM, red in V.
2. Turn the dial to DC volts: V with a straight line.
3. Black probe on –, red on +. Read the display.
4. A minus sign means they're swapped: nothing breaks.

SparkFun: how to use a multimeter
learn.sparkfun.com

A First light

At home, before anything is cut. One module on the Pi: this decides whether the plan holds.

ABOUT 2 H



01

One module, one test

1. Fit the Active Cooler on the Pi 5, then the tall header, then the bonnet on top. The bonnet lies **over the Pi and the cooler**, never off the side: turned round, pin 1 lands on pin 40.
2. Connect 12 V to **+Vin** and **−Vin** of the Mean Well: the powerbank's car socket, through a car plug with bare wires.
3. Switch on and measure **+V** and **−V**: it must read 5.0–5.2 V.
4. Switch off. Connect the module's power lead to +V and −V.
5. Connect the ribbon from **Port 1** to the module's input: the socket at the tail of the → arrow printed on its back.
6. Power the Pi from its own USB-C supply.
7. Run Adafruit Piomatter for one 64 × 32 panel, 1/16 scan, pinout **Active3**, colours RGB, at low brightness (settings: BUILD.md §10.2).

How to: Measure a voltage · Strip and crimp a wire end

✓ This passed on the bench on 4 Nov 2026 with one module. If a later batch is scrambled or flickers, stop and try the fallback: Pi 4 2 GB with the hzeller library. The 13-point checklist is in firmware/flushbuffer/README.md.

i Check the ↑ arrows before you judge a picture: a module on the desk with its input on the left is upside down.

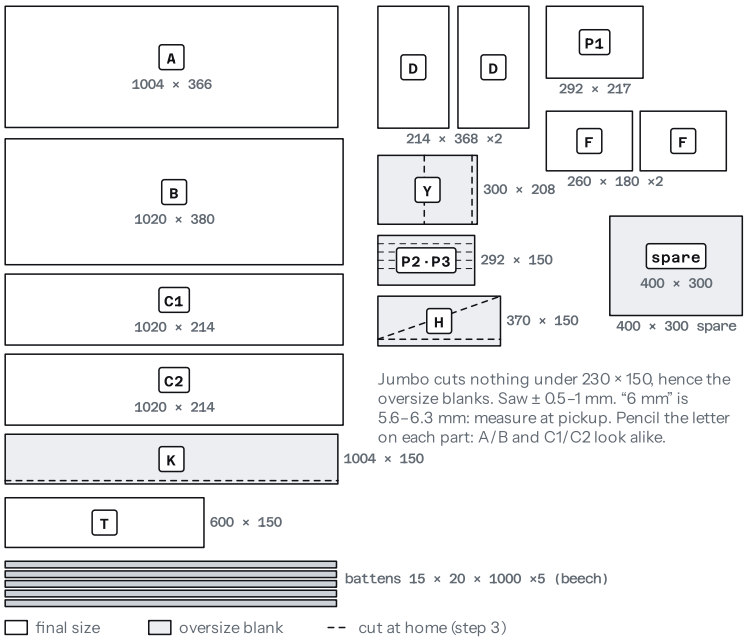
× Leave the bonnet's E4 / E8 pads open: these modules don't use the E line.

i Twisted bare wire is fine for this test. In the box every end gets a ferrule.

Workshop

B Saw, hole saw and metal work at the community-centre open workshop (GZ Hirzenbach: wood and metal, Tue 14–21, Sat 11–17; own tools and material allowed).

ONE WORKSHOP AFTERNOON



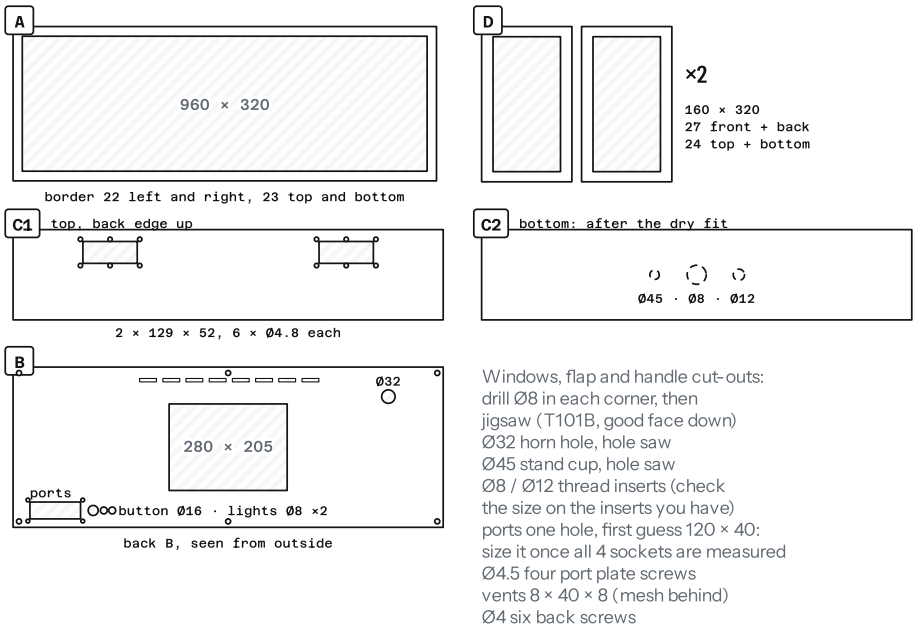
02

Get the parts

1. Collect the plywood from Jumbo: Oecoplan poplar 6 mm, already cut into rectangles.
2. Find the four oversize blanks: K, H, Y and the pocket walls. Jumbo cuts nothing under 230 × 150, so you finish these yourself.
3. Write each part’s letter on it in pencil.
4. Measure the real thickness of A, B and C1, and note it.
5. Add the five beech battens, 15 × 20 × 1000.

i Measure the real thickness at pickup: “6 mm” can be anything from 5.6 to 6.3. The outer joints get sanded flush later.

i Pencil each part’s letter on it as it comes off the saw.



Windows, flap and handle cut-outs:
drill Ø8 in each corner, then
jigsaw (T101B, good face down)
Ø32 horn hole, hole saw
Ø45 stand cup, hole saw
Ø8 / Ø12 thread inserts (check
the size on the inserts you have)
ports one hole, first guess 120 × 40:
size it once all 4 sockets are measured
Ø4.5 four port plate screws
vents 8 × 40 × 8 (mesh behind)
Ø4 six back screws

03

Cut the openings

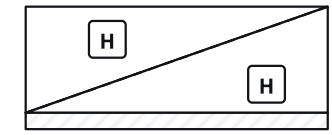
1. Tape the cut lines, good face down.
2. Drill Ø8 inside every corner of each opening.
3. Saw from hole to hole: the window in A, a window in each D, the flap opening in B, the two handle cut-outs in C1.
4. File the corners clean.
5. Cut the horn hole in B with the Ø32 hole saw, drilling from both sides.
6. Cut **one rectangular hole for the port plate**, low on the left, with its four Ø4.5 corner holes: only once all four sockets are measured. If they aren't here yet, leave it and cut it before B goes to paint.
7. Drill the rest of B: the **Ø16 button**, the two **Ø8 lights**, the vents and the six back screws.
8. Leave the holes in the bottom C2 until after the dry fit.

How to: Cut a window with the jigsaw · Cut a round hole

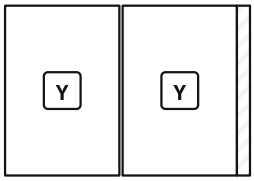
✓ Painter's tape on the cut line and the good face down: the T101B blade cuts on the up stroke, so the face down stays clean.

i About 120 × 40 for the hole is only a first guess. The plate is 10 mm larger on each side.

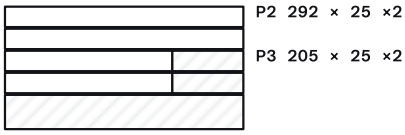


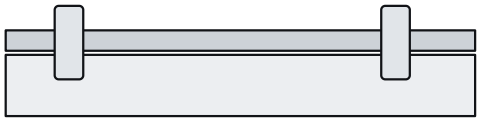


rip to 130, then corner to corner



2 × 140 × 208, glued as a pair





clamp a straight batten as a fence

T101B for straight rips, good face down. Cut beside the line on the offcut side, then sand to the line.

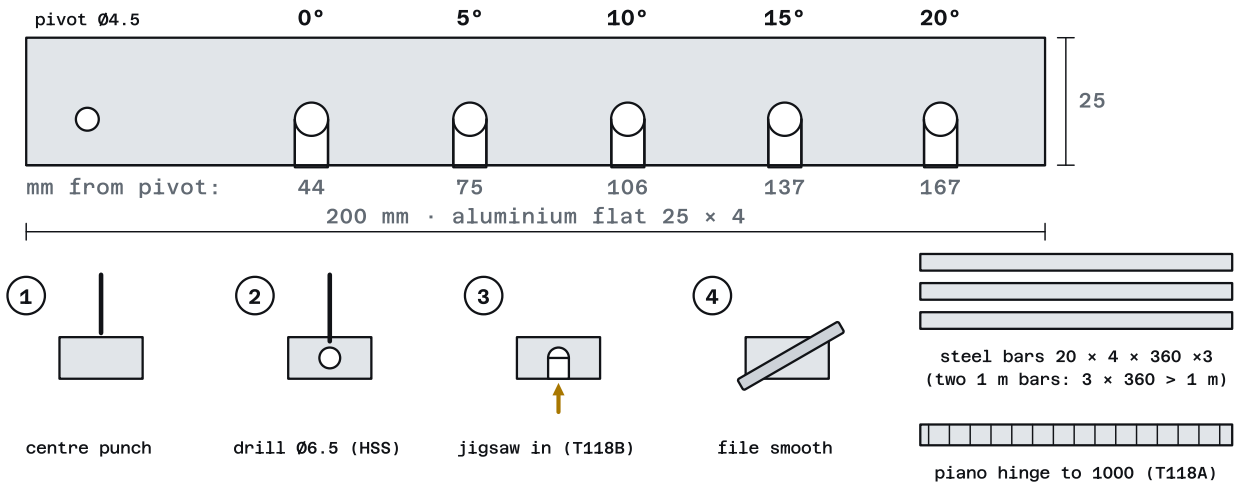
04

Rip the blanks

1. Cheeks H: rip the blank to 130, then saw corner to corner for two triangles.
2. Backing plates Y: cut two pieces 140 × 208.
3. Pocket walls: rip four strips 25 wide and cut two of them to 205.
4. Leave K at 150: the dry fit sets its depth.
5. Battens: cut two foot rails of 205, two tray posts of 362, and the blocks. Keep one batten spare.

How to: Saw a straight line

i A batten clamped as a fence keeps long rips straight. Cut beside the line on the offcut side, then sand to it.



Clamp the strip in the vise between two scraps of wood. Both stays together: tape them as a pair, drill through both. Notch spacing is from the hinge geometry; if your pivot or pin ends up elsewhere, mark the notches on the real box (step 10).

05

Make the metal parts

1. Cut two stays, 200 long, from the 25 × 4 aluminium.
2. Drill a Ø4.5 pivot hole 12 mm from one end of each.
3. Mark five notches 44, 75, 106, 137, 167 mm from the pivot (0, 5, 10, 15 and 20°), 9 mm up from the lower edge.
4. At each mark drill Ø6.5, cut in from the edge with the T118B blade on both sides of the hole, and file the slot smooth.
5. Cut three steel bars of 360 from the 20 × 4 flat.
6. Cut the piano hinge to 1000 (T118A, or sandwich it in plywood and use T118B).
7. Deburr every cut with a file.

How to: Cut and drill metal

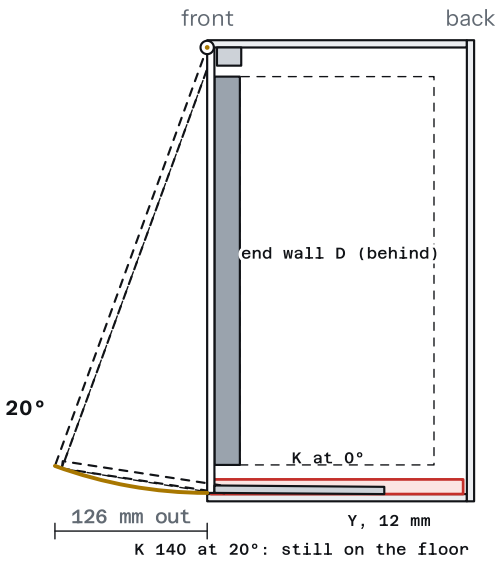
i The bars are 360, not 368: the front frame they sit on is only 366 tall. Three of them need two 1 m bars.

i Tape both stays together and drill through both: the notches then match.

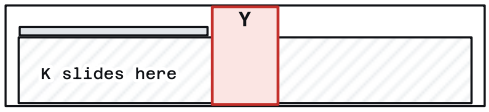
Case

C Back home: gluing, fitting and painting. Ventilate when painting; keep sawdust away from where you will wire later.

ABOUT 8 H + PAINT CURE



floor seen from above (front down)



Both want the middle of the floor. Decide here, before gluing Y:

1 Y towards the back, K rides over it on two thin runners
2 Y and the stand cup off K's path

- A swings free: 2 mm each side, 1 mm top and bottom
- K 140 mm: on the floor at 20° too, back edge 146 from the front at 0°

06

Dry fit

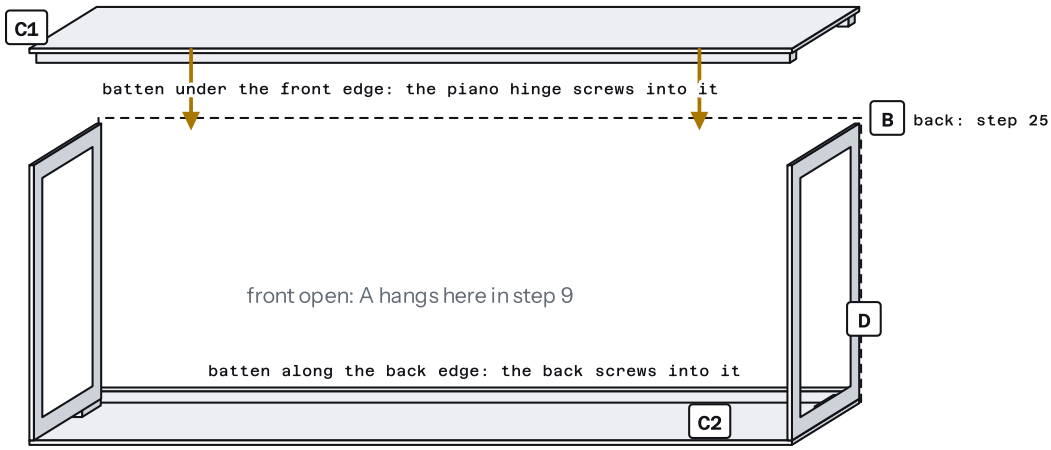
1. Tape the box together without glue: bottom, both ends, top and back.
2. Hold A in the front opening, with tape along its top edge as a hinge.
3. Swing it: it needs 2 mm each side and 1 mm top and bottom.
4. Hold the K blank against A's bottom edge and swing to 0° and to 20°.
5. Stand the tray upright on its posts and swing again: K must clear it.
6. Decide where Y goes, mark it, and rip K to its final depth.
7. Take everything apart and mark where each piece goes.

? **Open point.** The bottom cap K and the backing plates Y both want the middle of the floor: K slides over the front 14 cm, Y sits under the stand cup. Decide here, before gluing Y: **(1)** Y towards the back, with K riding over it on two thin runners, or **(2)** move Y and the stand cup off K's path. Check K's sweep at 0° and at 20°.

i At 20° the bottom of A stands 126 mm out. A 140 mm cap still reaches the floor (up to about 22°), so rip K to about 140 from the 150 blank. At 0° its back edge lies 146 mm behind the front: only 68 mm of floor is left behind it.

! **Check before you rely on it.** Hold the empty tray upright behind K (next to the two posts) and tilt from 0° to 20°: K must clear it. Check that the ribbons reach the modules over the tray's top edge with 12 cm of slack; longer HUB75 ribbons are cheap if not.





07

Glue the shell

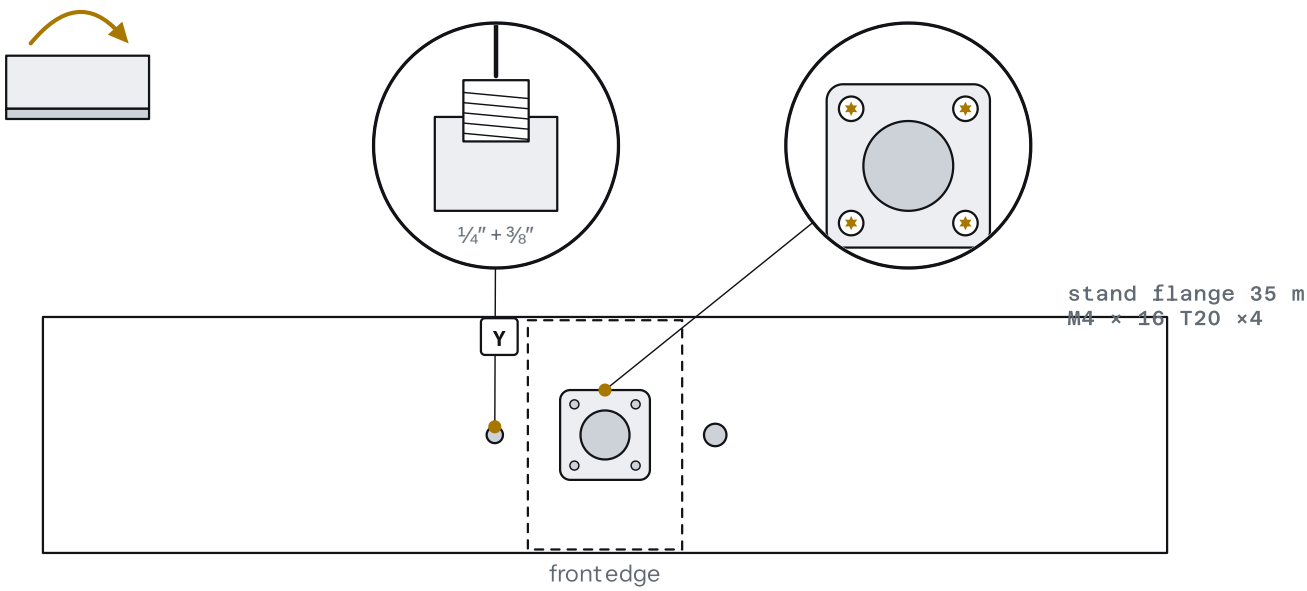
1. Spread D3 glue, and join the bottom C2, both ends D and the top C1.
2. Fit the batten under the top along its front edge, 20 mm face on the panel: the piano hinge screws into it.
3. Fit the batten on the bottom along the back edge: the back panel screws into it.
4. Add blocks from offcuts at the top back corners and middle, and in the inside corners. Keep them 40 mm back from the front.
5. Pre-drill Ø2.5 and screw through the panels into the battens.
6. Tape every 15 cm along the long joints, clamp where you can reach, and check it's square.
7. Leave it to dry overnight.

How to: Glue and tape a joint · Pre-drill and screw

i Pre-drill Ø2.5 for every 3.5 mm screw: beech is hard.

i The glue stays open about 10 minutes: one joint at a time.

✓ Before it sets: the front opening is 1008 × 368 and square (equal diagonals).



Y: two layers glued (12 mm), inside, under the flange and the threads.
Where Y and the cup sit follows the decision in step 5.

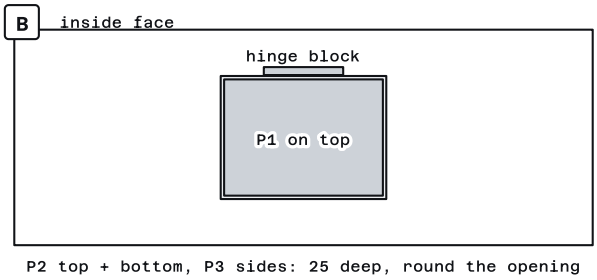
08

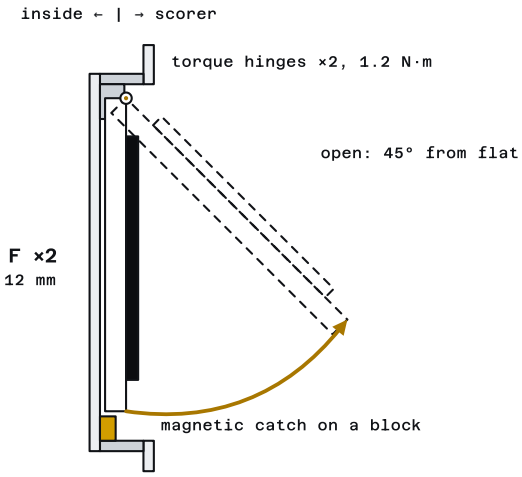
Stand cup and threads

1. Drill the bottom where the dry fit put Y: Ø45 for the stand cup.
2. Drill the holes for the two thread inserts, the size the inserts ask for.
3. Glue the two Y layers together, then glue them inside the bottom.
4. Fit the 35 mm stand flange with four M4 × 16.
5. Screw the ¼″-20 and ⅜″-16 inserts in from below.

How to: Cut a round hole

i The 6 mm bottom alone is too thin for bolts and inserts: they need the 12 mm plate.





1 Glue P2 and P3 round the opening, P1 on top: a closed box 25 mm deep, open to the back.
2 Glue the two flap layers F (12 mm). Trim to the real screen; 10 mm clear all round.
3 Hinge block under the top wall, two torque hinges at the flap's top edge.
4 Catch on a block at the bottom; a steel washer on the flap if there is no counter-plate.
5 A slot through F and P1 for the display ribbon.

09

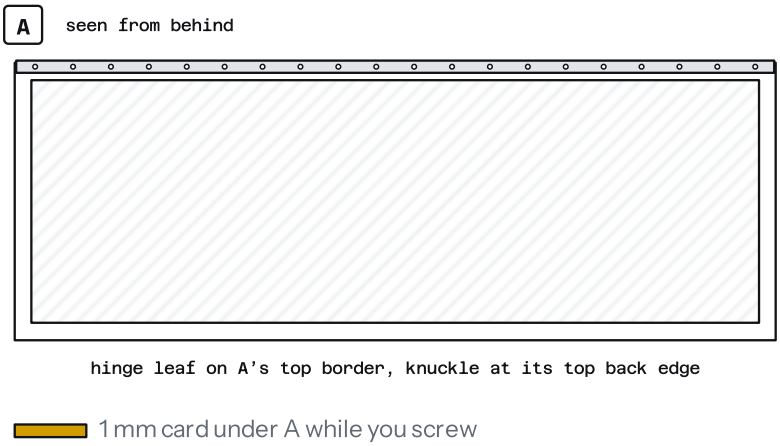
Screen pocket and flap

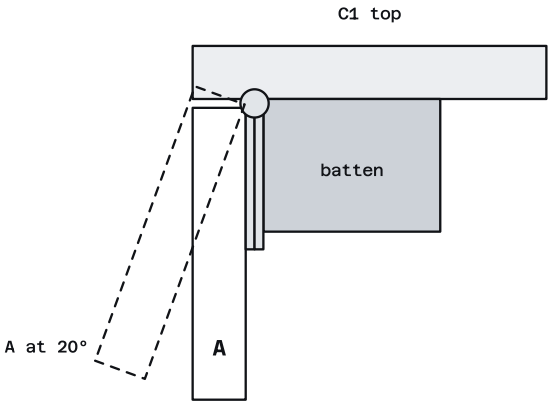
1. Glue the pocket walls P2 and P3 round the flap opening, on the inside of B.
2. Glue the pocket back P1 on top: this makes a box 25 mm deep.
3. Glue the two flap layers F together (12 mm).
4. Trim the flap to the real screen, 10 mm clear of the walls all round.
5. Fit two torque hinges along the flap's top edge, into a block under the pocket's top wall.
6. Fit the magnetic catch on a block at the bottom.
7. Cut a slot through F and P1 for the display ribbon.

How to: Glue and tape a joint

i The torque hinges hold the flap at any angle; it swings out like an awning from closed to about 75° open. Fit a stop there, for example a short cord from the flap to the pocket.

i Catches screw into blocks or battens, never into 6 mm plywood; pre-drill 2 mm. No counter-plate: a steel washer.





Keep the 3 mm round on A’s top front edge: the hinge sits on the inside, and at 20° that corner swings up about 2 mm into the top.

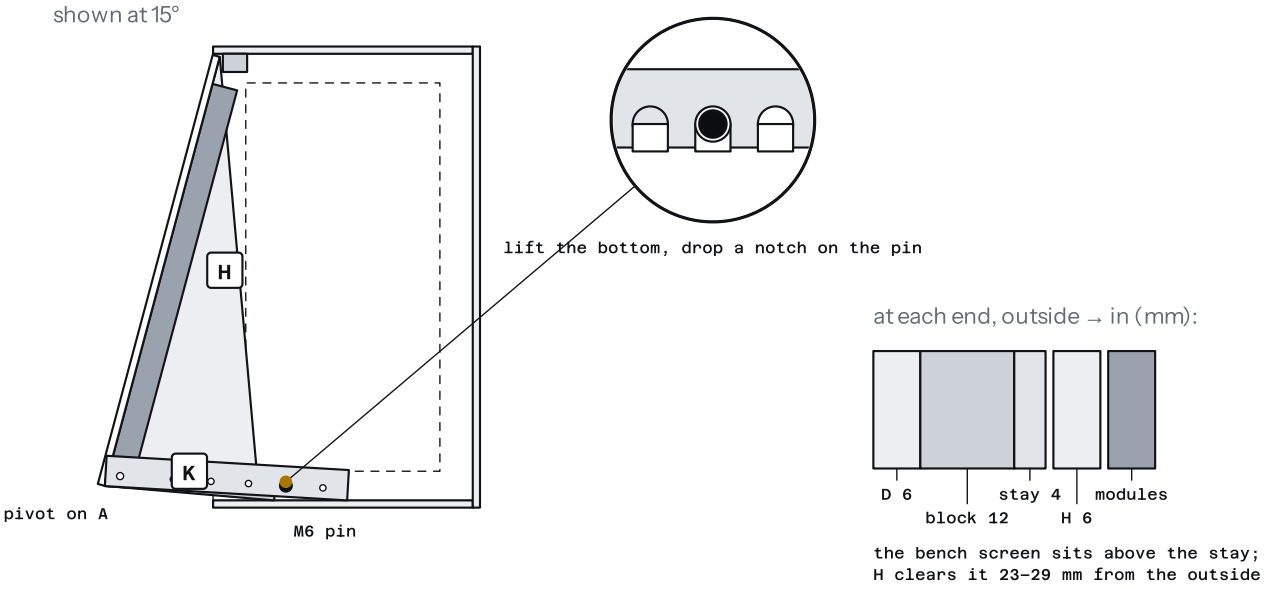
10

Hang the front frame

1. Lay A face-down.
2. Screw one hinge leaf along its top border, with the knuckle at A’s top back edge.
3. Put A in the opening with a 0.5 mm card under it and 2 mm on each side. 1.5 mm stays free under the top.
4. Screw the other leaf into the batten under the top.
5. Leave it for now: it comes off again for painting, but the screw holes are set.

How to: Pre-drill and screw

i Pre-drill 2 mm for the hinge screws.



11

Cap, cheeks and stays

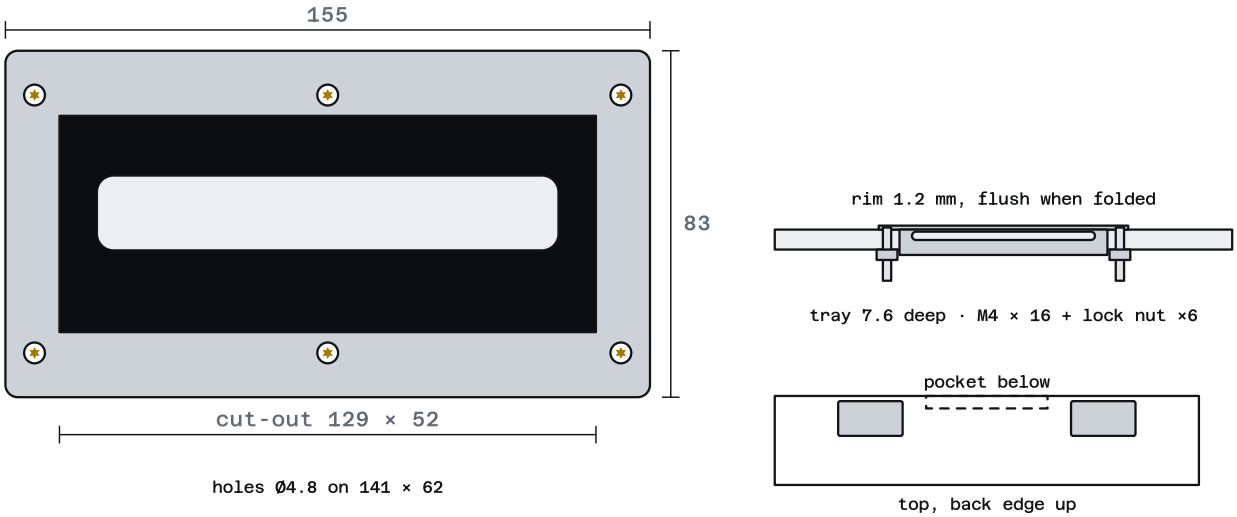
1. Glue each cheek H to a strip of batten on the back of A's end border, just inboard of the bench screen.
2. Hang K from A's bottom inside edge, so its back edge slides on the floor.
3. Fix a block on A's back at each bottom corner and fit the stay's pivot: a 3.5 × 30 screw with a washer, not tight.
4. Through each end wall's bottom border, 60 mm behind the front and 12 mm above the floor, fit the pin: an M6 × 30 bolt through a 12 mm block, with a nylon lock nut.
5. Lift A's bottom and drop a notch on the pin.
6. Check 0, 5, 10, 15 and 20° with the speed square.

How to: Cut and drill metal · Pre-drill and screw

i K hangs from A's bottom edge on three 50 mm butt hinges (3 × 12 screws; glue a 6 mm offcut doubler under each leaf on K and on A first, or the screws come through). Cut K's depth at the dry fit, about 140 from the 150 blank: a shallower cap no longer reaches the floor at 20°.

? **Open point.** The stay, its pin and the cheek share the 22 mm strip at each end of A. If the cheek touches the pin's nut, trim the cheek's bottom back corner: K closes that part of the gap anyway.

✓ Teeth don't creep when a ball hits the front.



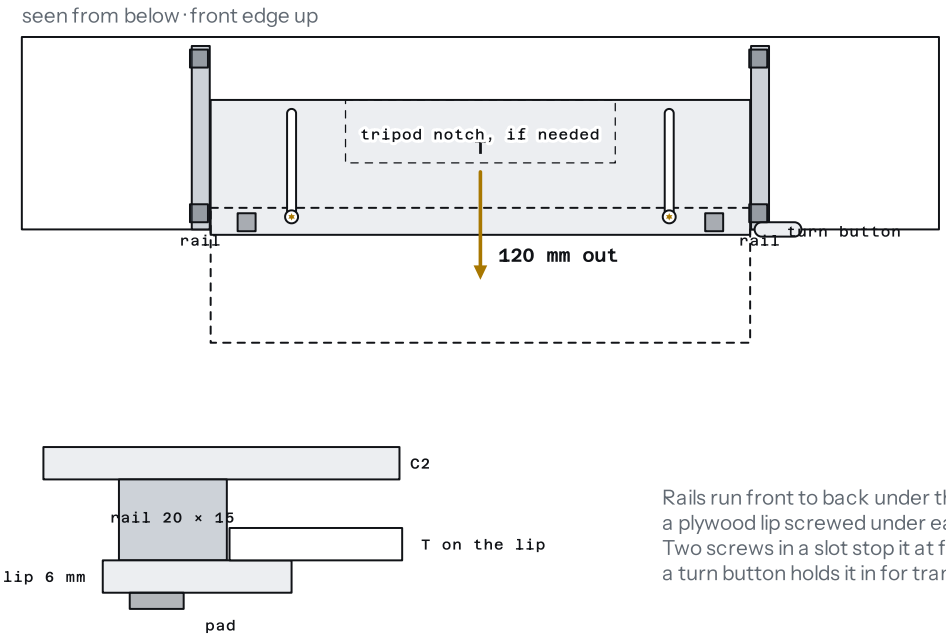
Near the back edge, left and right: clear of the flap pocket (under the middle of the back) and the hinge batten (along the front).

12

Handles

1. Drop each K16 handle into its cut-out from above, the rim flat on the top.
2. Push six M4 × 16 pan-head Torx screws through the rim.
3. Tighten a lock nut on each, inside the box.

i The tray sits 7.6 mm deep under the top: check it clears the blocks and the pocket.



Rails run front to back under the bottom; a plywood lip screwed under each holds T. Two screws in a slot stop it at full travel; a turn button holds it in for transport.

13

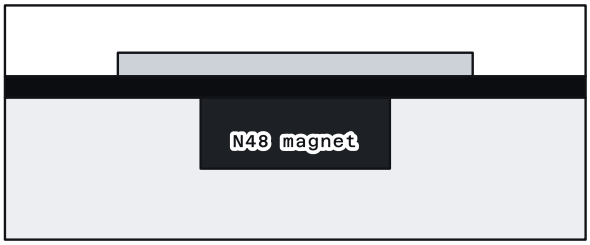
Pull-out foot

1. Glue two batten rails, 205 long, under the bottom from front to back, 602 apart inside.
2. Screw a strip of plywood offcut under each, overhanging 12 mm inward: the foot rides on these lips like a drawer.
3. Slide the foot T in. Pulled out 12 cm, it turns the 22 cm base into 34 cm.
4. Fit two screws in a slot to stop it at full travel.
5. Fit a turn button on the back end of a rail to hold it in.

How to: Glue and tape a joint · Pre-drill and screw

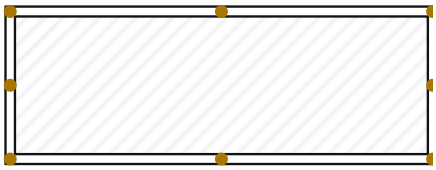
i With the foot in, check that a tripod plate still reaches the threads; if not, cut a notch in T's front edge.

section at one magnet (enlarged, to scale)



cover, 3 mm
M5 washer, sunk 1 mm
felt, 0.8–1 mm
magnet Ø8 × 3, flush
box border, 6 mm

front A: 8 magnets



each end D: 4



flap lid: 4



Only 10 mm border: smaller washers, in the corners. Check at dry fit.

20 magnets in all

- Forstner bit. A spade bit's spur goes right through 3 mm.
- Tape on the bit as a depth stop: 3 mm in the box, 1 mm in the cover.
- Drill before the paint; glue magnets and washers in after it.
- Marking the covers: a dab of paint on each magnet, press the cover on.

14

Pockets for the covers

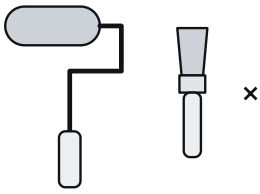
1. Put tape on the drill bit as a depth stop.
2. Drill the magnet pockets, Ø8 × 3 mm, into the borders of A, both D and around the flap on B.
3. Lay each cover on and mark the washer positions.
4. Drill the washer pockets about 1 mm deep in the 3 mm covers.
5. Do this now, before the paint. Magnets, washers and felt go in once the paint is dry.

How to: Sink a magnet or washer pocket

× No 4 mm felt: it holds the magnet too far away.

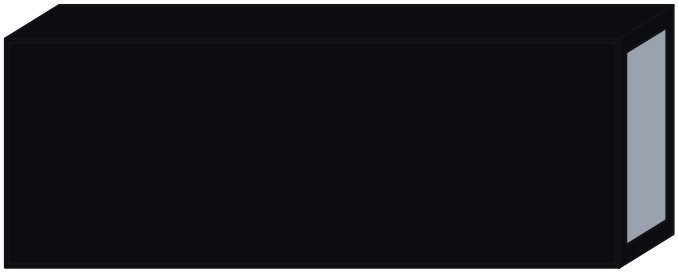
i Washers are plain steel, so the magnets have no polarity to get wrong.

Hinge, stays, handles and catches off first; label them.



×2

coats, dry betwee



box, inside and out: matte black



front face of A: gloss black

15

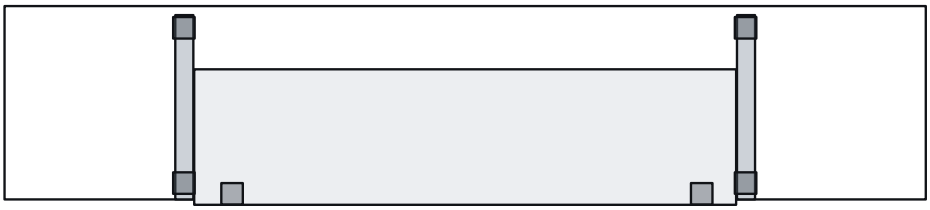
Paint

1. Take the hinge, stays, handles and catches off, and label them.
2. Round every outside edge about 3 mm with sandpaper: P120, then P180.
3. Fill the grain on A's front face, and sand.
4. Paint everything matte black, two coats, inside too, so nothing reflects behind the LEDs.
5. Paint A's front face gloss black.
6. Let it cure. Then glue in the magnets, washers and felt.

How to: Spray paint

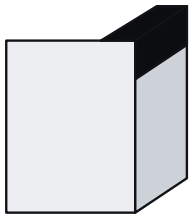
- i Keep the 3 mm round on A's top front edge: at 20° that corner swings up about 2 mm into the top.
- i Paint at the workshop if they allow it, or on a balcony on a drop sheet. Optional: a 2K matte clear on top against splashes.

seen from below



4 under the rails 2 under the foot’s back end

2 spare · press 30 s each



rubber cap, front corners
only as far as the end’s 6 mm edge:
the front frame must swing past it

only after the paint has fully cured (see the tin)

16

Pads and corner caps

1. Wait until the paint has fully cured.
2. Put the hardware back.
3. Stick four clear rubber pads under the rails and two under the foot’s back end.
4. Trim the black corner caps so they cover only the ends’ 6 mm edge: the front frame must swing past them.
5. Fit them on the four front corners.

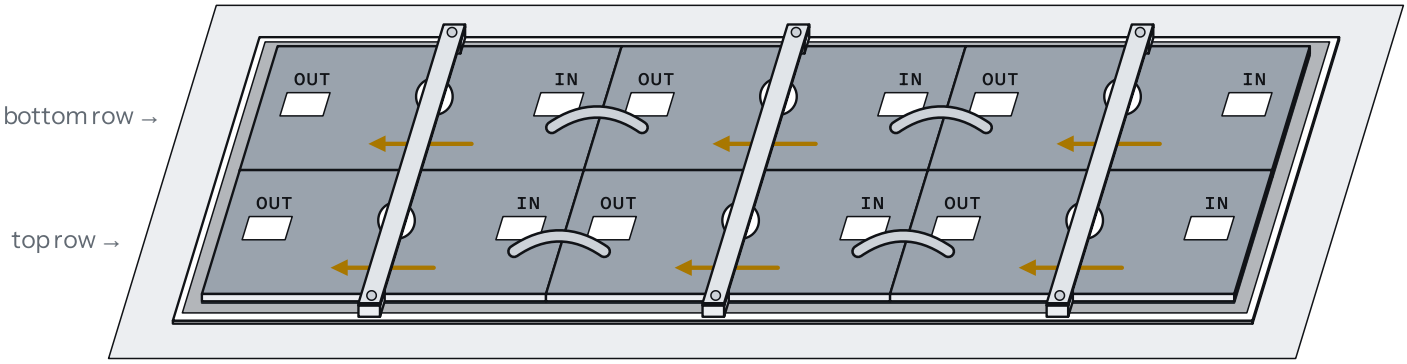
D Screens

The front frame comes off its hinge once more for the modules.

ABOUT 2 H

A face-down on a blanket

Bars sit on the magnets, screwed to blocks on A's top and bottom border, packed to the magnet height.



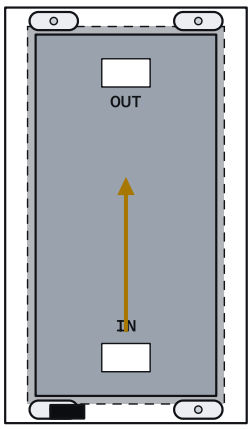
17

Modules on the front frame

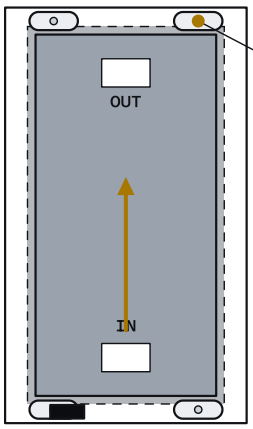
1. Unscrew the hinge from the batten (it stays on A) and lay A face-down on a blanket.
2. Peel the inner film off the smoked sheet and lay it over the window.
3. Lay six modules on top, LEDs down, all arrows the same way.
4. Connect the short ribbons: 3 → 2 → 1 and 6 → 5 → 4.
5. Fit four M3 magnetic screws in each module.
6. Glue blocks on A's top and bottom border, packed to the magnet height.
7. Screw the three steel bars onto the blocks: the magnets snap on, so a module swaps out in seconds.

✓ Leave the IN of modules 3 and 6 free: the driver plugs in there. The module nearest the Pi shows the last slot of its row.

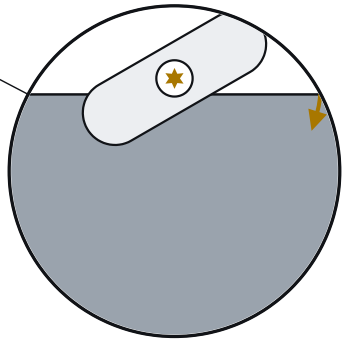
i Keep the sheet's outer film on until the very end.



left end, from inside
pin block



right end, from inside
pin block



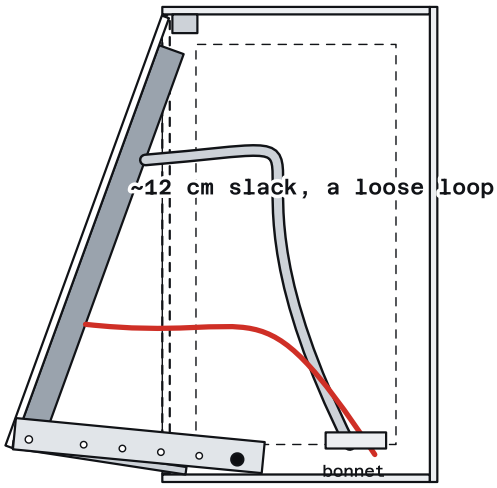
Smoked sheet 172 × 332 first, then the module, arrows up. 8 turn buttons, 4 per end. Keep the pin block clear.

18

Bench screens

1. Lay one smoked sheet behind each end window.
2. Put one module on it, arrows up.
3. Twist four turn buttons over the module’s edges.

i The stay’s pin block sits just under the bench screen: keep the bottom border free.



Screw the hinge leaf back into the batten (same holes). Then plug the ribbons and power leads with about 12 cm of slack: tilt to 20° and back with everything plugged in. Nothing may pull tight or catch on a cheek.

bench screens don't move

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Hang it again

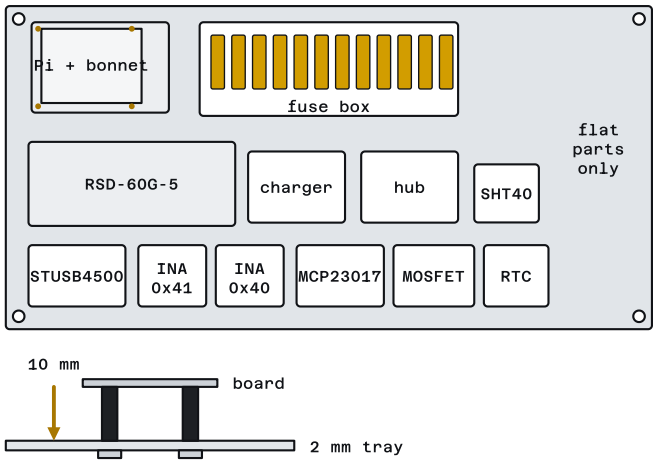
1. Screw the hinge back into the batten, same holes.
2. Plug in the ribbons and power leads, with about 12 cm of slack.
3. Tilt to 20° and back with everything plugged in: nothing may pull tight or catch on a cheek.

E Electronics

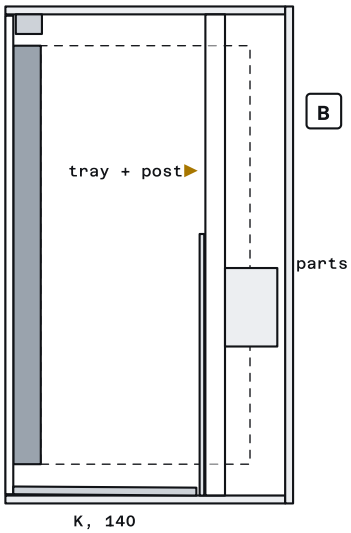
Low voltage only: nothing in here can give you a shock. Solder at home, with a window open.

ABOUT 6 H

the tray: 400 × 200, 2 mm aluminium, seen from the back



side view, to scale



- Ø3 for M2.5, deburr. No joint touches the metal.
- Converter and fuse box flat on the aluminium: it cools them.

20

Build the tray

1. Lay all the parts on the aluminium sheet, in the rows shown.
2. Mark the holes through the parts and drill: Ø3 for M2.5. Deburr.
3. Screw the converter and the fuse box flat on the metal.
4. Put everything else on standoffs.
5. Fix the two posts, 362 long, between floor and top, 150 mm behind the front.
6. Screw the tray to the posts with four screws, its parts facing the back.

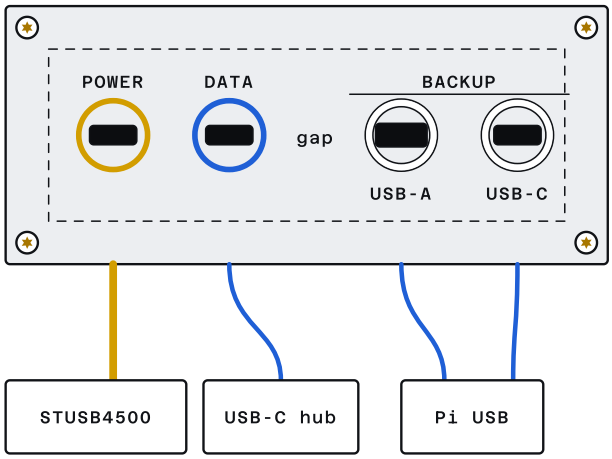
How to: Cut and drill metal

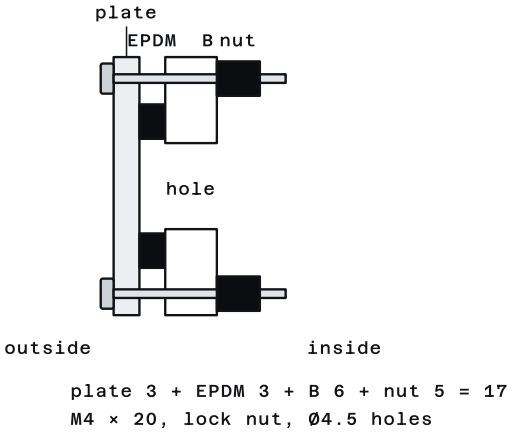
i Leave 15 cm of slack on everything that goes to the back panel, so it can come off.

i Pi at the top, nearest the back: its Wi-Fi is the only link out, and it can't get through the modules.

! **Check before you rely on it.** Only 64 mm stands free from tray to back panel, and 39 mm where the flap pocket is. Keep the tall parts away from the pocket, and check the Pi + bonnet + ribbon stack at the dry fit.

plate from outside: first guess 140 × 60, hole 120 × 40 dashed





Inside, everything stays separate: the plate only holds the sockets.

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Port plate

1. Measure all four sockets: flange, cut-out, fixings and how far each reaches inside.
2. Fix the size: the hole takes all four, and the plate is 10 mm larger on each side. About 120 × 40 for the hole and 140 × 60 for the plate is only a first guess.
3. Cut each socket its own opening in 2–3 mm black plastic, left to right: POWER, DATA, a gap, then BACKUP USB-A and USB-C.
4. Fit the sockets, each with its own fixings, then the rings and labels: gold POWER, blue DATA, white BACKUP.
5. Stick EPDM round the hole, on the back of the plate. Cut it at the corners so the screws don’t pierce it.
6. Bolt the plate on with four M4 × 20 screws and lock nuts, 5 mm in from its edges.
7. Inside, keep everything separate: POWER to the STUSB4500, DATA to the USB-C hub, BACKUP on two cables to the Pi’s USB ports.

✓ A socket cut wrong costs a new plate, not a new back panel.

i Leave 15 cm of slack: the plate stays on the back panel and its cables stay plugged in when the panel comes off. Check that the 30 cm cables reach.

i A thin bead of black silicone also seals, but it has to be cut and redone each time the plate comes off.

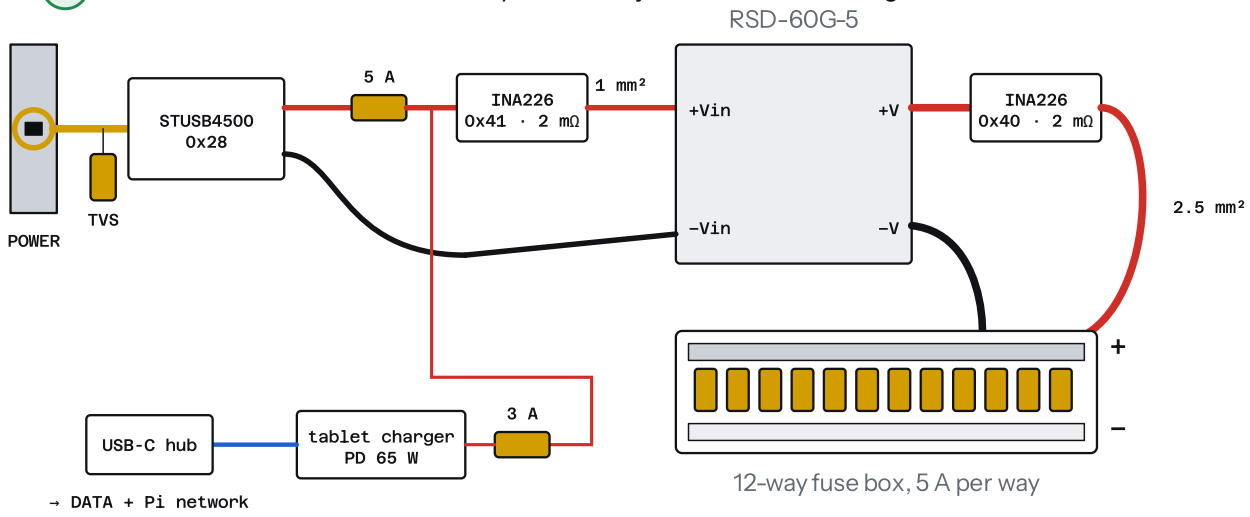
× Aluminium only after the meter check in BUILD.md §8.9: a metal plate joins the sockets’ shells.

× BACKUP is for a stick or SD reader only. A charger or a tablet doesn’t work there.

! **Check before you rely on it.** With a charger in the BACKUP USB-C socket, no 5 V may appear at its USB-A plug. Try a USB-C stick both ways round.

! **Check before you rely on it.** Only 64 mm is free between the back panel and the tray. Check each socket and its cable bend fit, with the button and lights beside the plate.

red = +, black = −, a ferrule on every end; −Vin joined to −V (shared ground)



never swap + and −; STUSB4500 set to “output only above 5 V”

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Power path

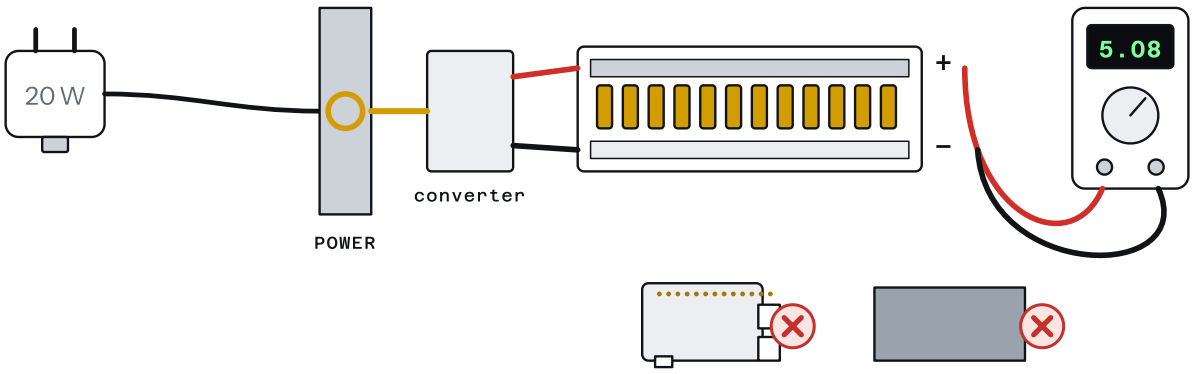
1. Wire POWER socket → STUSB4500, with the TVS diode across + and − at the socket.
2. Wire STUSB4500 → 5 A fuse → INA226 (0x41, A0 bridged) → converter, in 1 mm².
3. Join −Vin to −V on the converter.
4. Wire converter → INA226 (0x40) → fuse box, in 2.5 mm².
5. After the 5 A fuse, wire a 3 A fuse → tablet charger → the USB-C hub’s charging input.
6. Put a ferrule on every wire end.

How to: Strip and crimp a wire end · Solder a joint

- Set the STUSB4500 to “output only above 5 V”: a 5 V-only source leaves the box off instead of half-booting it.
- Join the converter’s −Vin to −V: the USB-C board talks to the Pi and needs a shared ground.
- The Pi takes its 5 V on the header pins from its own fuse; its USB-C socket stays empty.

5.0–5.2 V → next step

0 V or −5 V → unplug, check + / −



Pi and modules not connected yet

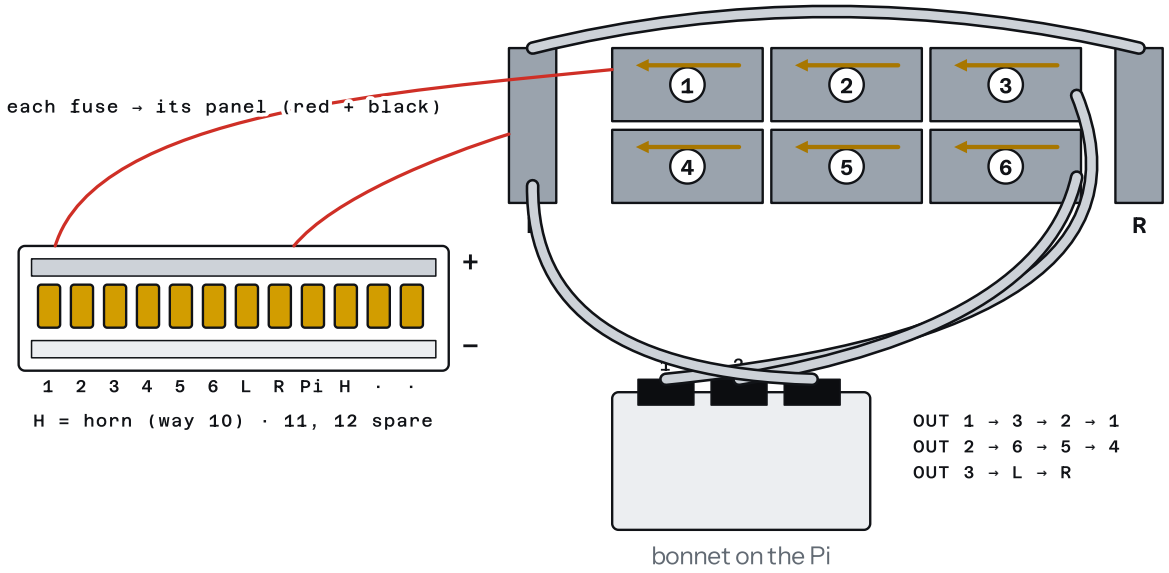
23

First power-up

1. Check nothing else is connected.
2. Plug a small USB-C phone charger into POWER.
3. Measure at the fuse box: it must read 5.0–5.2 V.

How to: Measure a voltage

i A weak charger is the safety net: it cannot push enough current to damage anything.



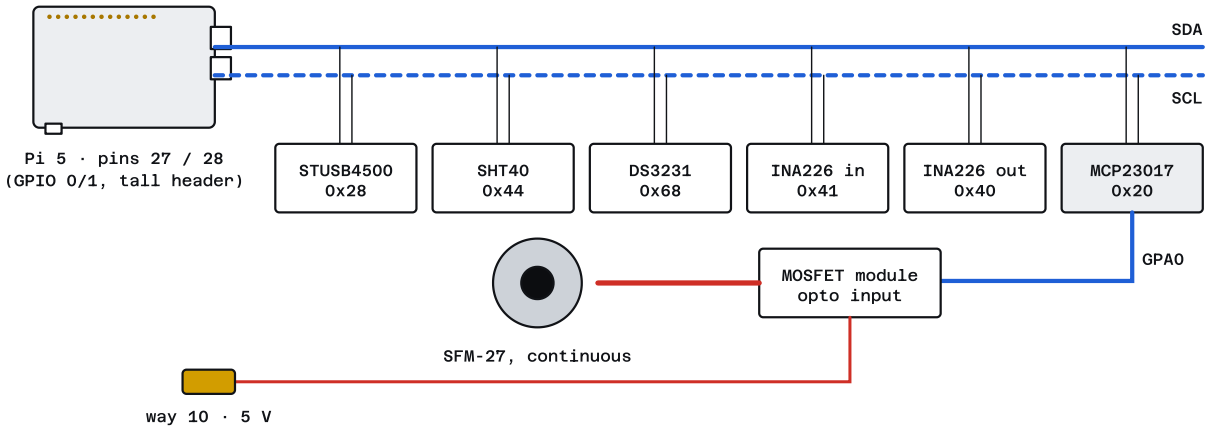
24

LED power and data

1. Connect each of the six modules to its own fuse way: 1–6.
2. Connect the bench modules to ways 7 and 8, and the Pi to way 9.
3. Keep way 10 for the horn. 11 and 12 are spare.
4. Connect the ribbon from OUT 1 to module 3, and OUT 2 to module 6.
5. Connect OUT 3 to the left bench module, and on to the right one.

How to: Strip and crimp a wire end

i Panel power leads that don't reach: extend with 1.5 mm².




```
i2cset -y <bus> 0x20 0x14 0x00 output latch low
i2cset -y <bus> 0x20 0x00 0x80 GPA0-6 = outputs
i2cset -y <bus> 0x20 0x14 0x01 horn on · 0x14 0x00 off
```

MCP23017: A0–A2 to GND, RESET to 3.3 V. INA226 never at 0x44 (that is the SHT40). DS3231: CR2032 in, charging resistor off. Never a PCF8574 for the horn.

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I²C parts and the horn

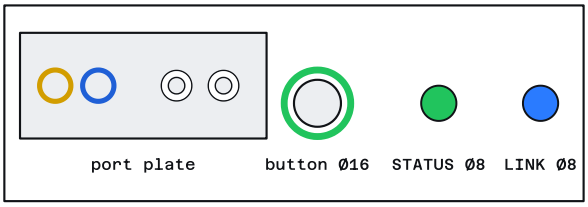
1. Join the USB-C board, the clock, the temperature sensor, both current sensors and the MCP23017 on two wires.
2. Connect those wires to pins 27 and 28 (GPIO 0 and 1) of the tall header.
3. Connect MCP23017 pin GPA0 to the input of the isolated MOSFET module.
4. Wire fuse way 10 through the MOSFET module to the buzzer.
5. Screw the buzzer on from inside, over the Ø32 hole.

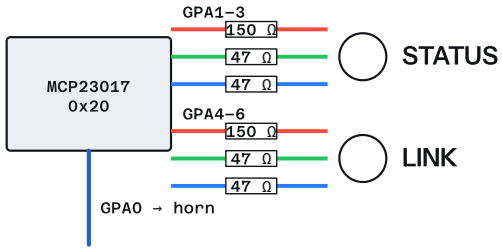
How to: Solder a joint

! **Check before you rely on it.** Chain 3 uses GPIO 2, 3 and 14, which is why I²C moved to pins 27/28. Confirm this still holds on the Pi 5 with the bonnet before you wire it.

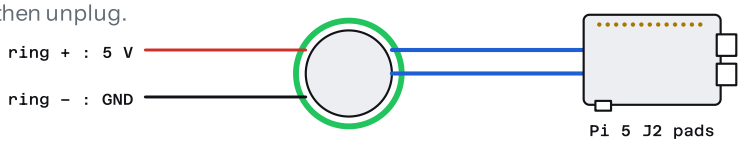
i The MCP23017 starts with every pin an input, so the horn stays off at boot.

back panel B, seen from outside





The button's green ring is the POWER light.
Hold the button 2 s: the Pi shuts down, then unplug.



what they say

- POWER ring on: power is there
- STATUS yellow, blinking: starting
- STATUS green: ready
- STATUS red, blinking: error: read the screen
- LINK blue: tablet on the cable
- LINK yellow: Wi-Fi, no internet
- LINK green: online
- LINK red, blinking: live results failing

26

Button and lights

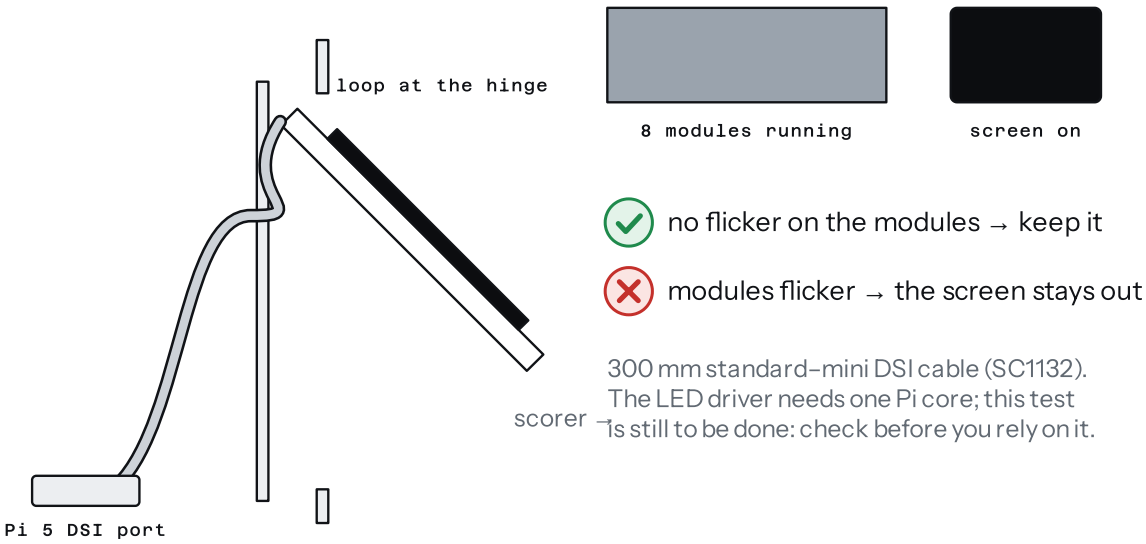
1. Press the 16 mm button into its Ø16 hole.
2. Press the two LED holders into their Ø8 holes, beside the port plate.
3. Solder a 2-pin header on the Pi 5's J2 pads and connect the button's switch contacts to it.
4. Connect the green ring straight to 5 V: it shows power even when the Pi is dead.
5. Connect GPA1–3 to STATUS and GPA4–6 to LINK, through 150, 47 and 47 Ω.

How to: Solder a joint

i There is one button and it doesn't cut power: hold it 2 s and the Pi shuts down cleanly. To switch off, unplug.

i Until the software drives them, STATUS and LINK stay off: no light is better than a wrong one.

! **Check before you rely on it.** Green and blue may be dim at 3.3 V. Look at them on the bench before you close the box.

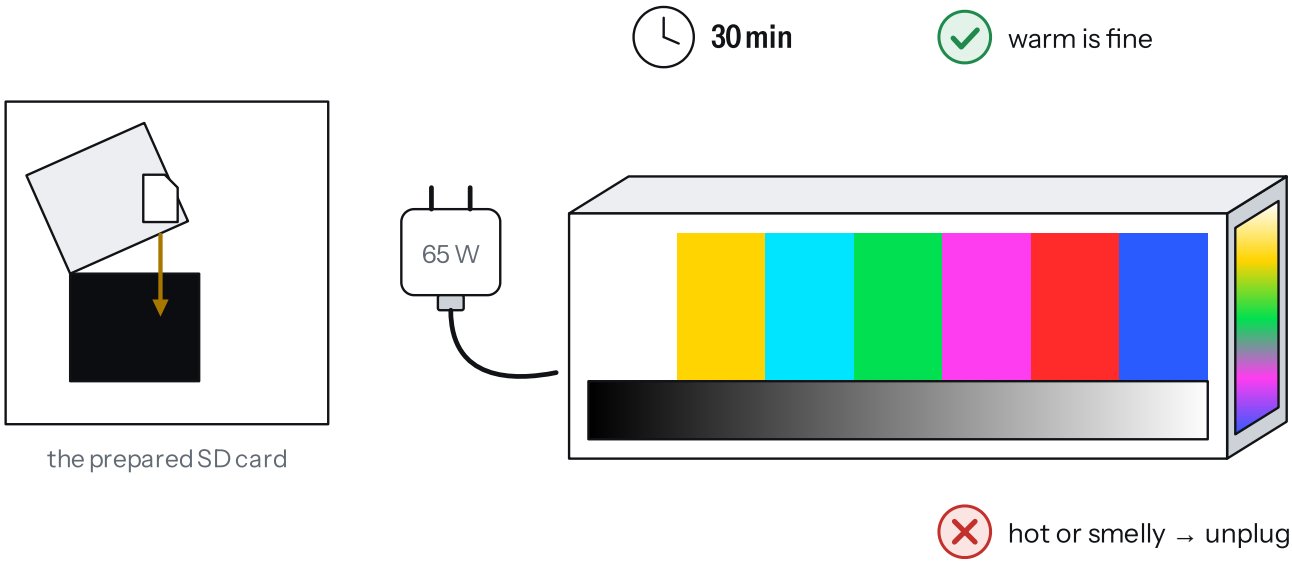


27

Control screen

1. Screw the screen onto the flap's outer face.
2. Pass its ribbon through the slot.
3. Run the 300 mm DSI cable to the Pi 5's display port, with a loose loop at the hinge so it flexes instead of folding.

! **Check before you rely on it.** Run all eight modules and the screen together. The LED driver needs one Pi core; if the modules flicker with the screen on, the screen stays out.



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Test

1. Leave the back off and slide in the prepared SD card.
2. Power up with a 65 W laptop charger.
3. POWER lights green at once. STATUS blinks yellow, then turns green.
4. LINK shows blue with the tablet on the cable, green when online.
5. Let the self-test run, then the test pattern on all eight modules, for 30 minutes.
6. Read the result on the diagnostics page (BUILD.md §6).

How to: Measure a voltage

× Before the first match the SD card must run read-only: then pulling the plug can't corrupt it. Unplugging is this board's off switch.

! **Check before you rely on it.** The software is written, but it has not run on the real chips yet: see the status table in BUILD.md §10.7.

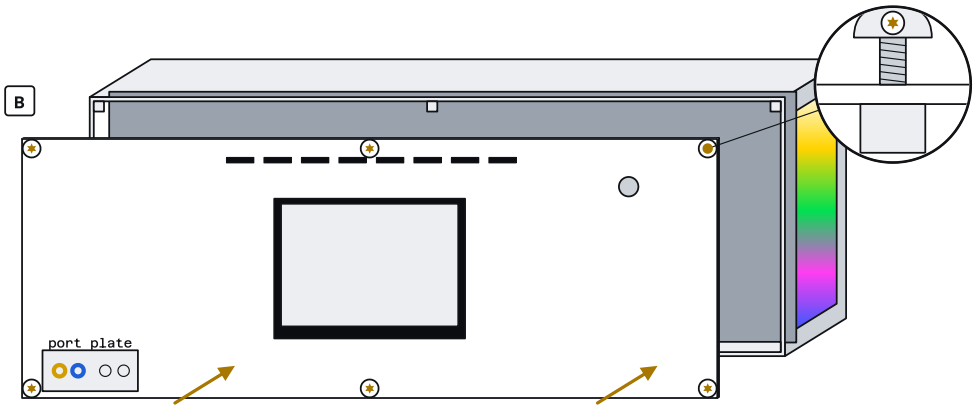
! **Check before you rely on it.** First light on the Pi 5 with Piomatter has passed, but the screen flicker test and the soak test are not done yet. If the Pi 5 path fails, a Pi 4 2 GB with the hzeller library is the fallback (BUILD.md).

F Finish

Ready for the hall.

ABOUT 1 H

×6 T20 3.5 × 16 into the batten and blocks

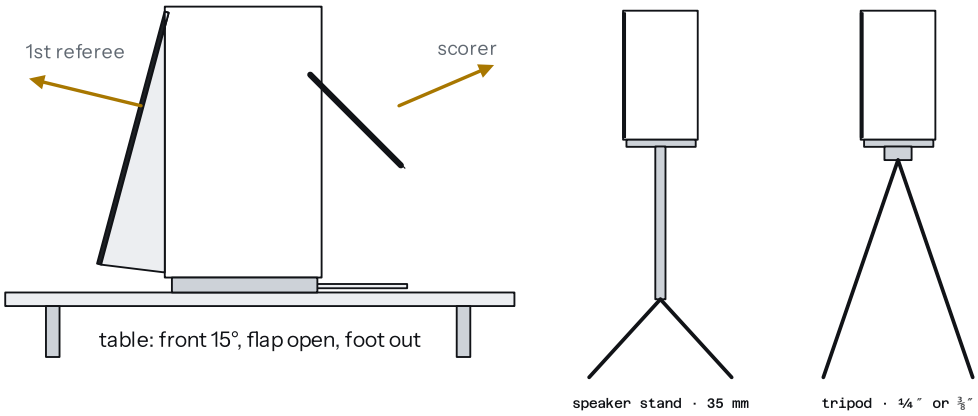


EPDM D-seal round B's rim, dust caps on all four port sockets.
Inside B: fuse map, spare fuses, spare SD. Lettering and the back label after first light.

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Close up

1. Put the fuse map, spare fuses and a spare SD card inside the back.
2. Fit the EPDM D-seal round the back panel's rim, and the brush seal at the front frame's gap.
3. Put the back panel on and drive in the six T20 screws. Turn buttons hold the rest.
4. Put dust caps on all four port sockets.
5. Stick on the POINT HUB lettering and the back label after first light, when the port plate and lights are final.



POWER ← any USB-C charger, 18–100 W (65 W+ for full brightness)

DATA ← tablet: scores + charging, no PIN on the cable

BACKUP ← USB stick or SD reader: no charger, no tablet

30

Set it up

1. **On a table:** lift the front frame's bottom and drop the stays into a notch, 0 to 20° toward the first referee.
2. Swing the flap open for the scorer and pull the foot out.
3. **On a speaker stand or tripod:** foot in, flap as needed. The 3 m USB-C cable keeps the tablet on the scorer's table.
4. Plug the charger into POWER.
5. Plug the tablet into DATA: one cable carries the scores and charges it, and no scorer PIN is needed on it.
6. Plug a USB stick or SD reader into BACKUP for a copy of the eScoresheet backups during the match.

✓ Any USB-C charger works from 18 W; 65 W or more gives full brightness and charges the tablet.

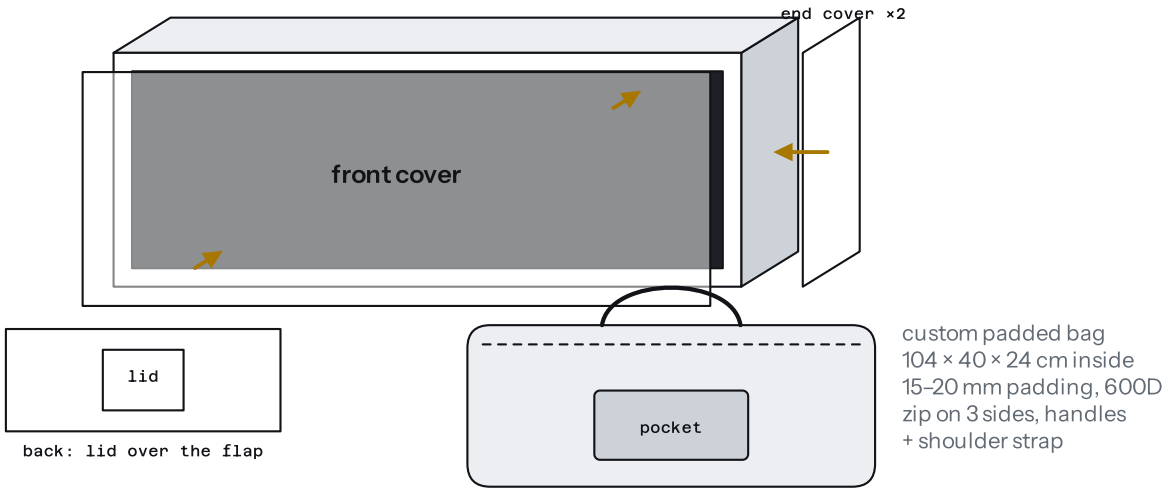
× Never plug the DATA cable into the school's network: it's for the tablet or a laptop only.

× Never in direct sun or by a sunny window: the box would overheat. It dims itself from 50 °C.

i Keep a flip scoreboard with digits of 12 cm or more in the kit: Swiss Volley requires one if the board fails.

i Coming in from a cold car: wait 45 minutes before switching on, or condensation can short the modules.

Four covers snap on with the magnets. All lie inside the bag.



custom padded bag
104 × 40 × 24 cm inside
15–20 mm padding, 600D
zip on 3 sides, handles
+ shoulder strap

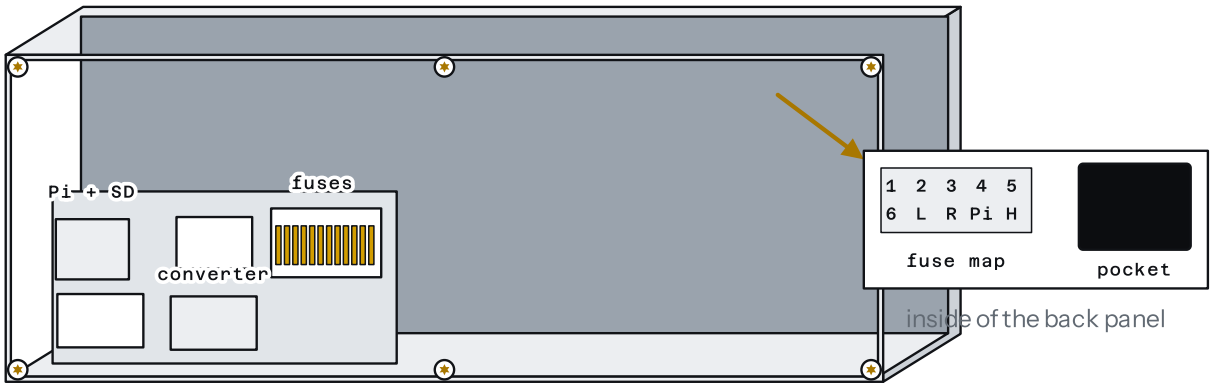
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Covers and bag

1. Snap on the front cover.
2. Snap on both end covers.
3. Snap on the lid over the flap.
4. Put the box in its padded bag, with chargers and cables in the outer pocket.
5. Order the bag once the box is built, from the real measurements.

i Keep the T20 key, the flip scoreboard and the 3 m USB-C cable in the outer pocket.

T20 key: six screws, lift off, unplug the screen ribbon



inside of the back panel

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Open it

1. Unplug the charger.
2. Undo the six T20 screws.
3. Ease the back panel off: the port plate stays on it, with its cables plugged in.
4. Unplug the screen ribbon at the Pi: the tray faces you and everything is in reach.

✓ Swapping the SD card or a fuse takes a minute.

i The back takes a T20 key on purpose: in a public hall, anything that opens by hand gets opened. Keep the key in the kit bag.

KSC Wiedikon · Point Hub board build manual, design D, draft of 4 November 2026.

The panel software comes ready on the SD card. Wiring, settings and safety: BUILD.md.